

Amelia Dobis

PHD CANDIDATE AT PRINCETON UNIVERSITY · PROGRAMMING LANGUAGES AND COMPILERS

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research interests: Hardware Language Design, Compiler Construction, Modal Logics, Formal Verification

Education

Princeton University

Princeton, NJ, USA

DOCTOR OF PHILOSOPHY, PHD IN COMPUTER SCIENCE

August 2024 - present

- Advisor: Mae Milano, Princeton PL/SNS groups; co-advised by Kevin Laeuer at Cornell University.
- Researching Programming Languages for Hardware-Software Co-Design and Verification.

Eidgenössische Technische Hochschule Zürich (ETH Zürich)

Zürich, ZH, Switzerland

MASTER OF SCIENCE, MSc. IN COMPUTER SCIENCE

August 2021 - April 2024

- Major: Secure and Reliable Systems; minor: Computer Graphics.
- Thesis: Formal Verification of Hardware using MLIR, advised by Kevin Laeuer (UC Berkeley) and Prof. Zhendong Su (ETH Zürich).

École Polytechnique Fédérale de Lausanne (EPFL)

Lausanne, VD, Switzerland

BACHELOR OF SCIENCE, BSc. IN COMPUTER SCIENCE

August 2016 - Septembre 2020

- Focus: Computer Engineering.
- Thesis: Accelerating Ray-tracing on FPGA (at the Processor Architecture Lab (LAP) EPFL).

Experience

Research Engineer

Paris, IDF, France (remote)

YosysHQ

June 2026 - Present

Part-time researcher on the PropertyIR project, focused on developing a general solution for gradually synthesizing SVA properties.

Doctoral Researcher

Princeton, NJ, USA

PRINCETON UNIVERSITY

August 2024 - Present

Working at the intersection of Programming Languages and Computer Architecture to create a unified language for embedded systems projects that can model and verify full hardware-software co-design systems.

Compiler Engineer

Berkeley, CA, USA

SiFive, Inc.

March 2024 - August 2024

Worked on both the CIRCT and Chisel teams on various formal verification projects. Specifically on extending temporal logic support in CIRCT and introducing a verification based Intermediate representation capable of supporting modular specifications.

Visiting Researcher

Berkeley, CA, USA

UNIVERSITY OF CALIFORNIA, BERKELEY

August 2023 - April 2024

Extended the CIRCT hardware compiler framework to support formal verification of linear temporal logic specifications.

Research Assistant

Lugano, TI, Switzerland

UNIVERSITÀ DELLA SVIZZERA ITALIANA (USI)

June 2023 - April 2024

Software Engineer for the SURE sustainable development project, where we developed a video game called “Ensured Energy”, illustrating the impacts of energy management decisions, as well as natural and societal disasters, on the Swiss energy grid.

Software Engineer

Lausanne, VD, Switzerland

GAMELAB LAUSANNE

March 2020 - March 2023

Game developer for the GameLab research group at the University of Lausanne in collaboration with Digital Kingdom and LeTemps across two projects named “Lausanne 1830” and “Quatre Apparts et un Confinement”.

Research Assistant

Copenhagen, Denmark

TECHNICAL UNIVERSITY OF DENMARK (DTU)

September 2020 - September 2021

Researched High-level hardware verification, through the Chiselverify project, which adds support in Chisel for Constraint Random Testing, Functional Coverage, Bus Functional Models and functional coverage driven mutation based fuzzing for digital circuits.

Publications

Encoding Purely Functional Languages in an RTL-based Compiler

AMELIA DOBIS*, GONGQI HUANG*, MAE MILANO

LATTE 2026

March 2026

Incremental Conversion of SVA Properties to Synthesizable Hardware

AMELIA DOBIS, FABIAN SCHUIKI, MAE MILANO

LATTE 2025

March 2025

Formal Verification of Hardware using MLIR

AMELIA DOBIS

Masters Thesis, ETH Zürich

April 2024

Verification of Chisel Hardware Designs with ChiselVerify Microprocessors and Microsystems, Volume 96

AMELIA DOBIS, KEVIN LAEUFER, HANS JAKOB DAMSGAARD, TJARK PETERSEN, KASPER HESSE, ENRICO TOLOTTO, SIMON THYE ANDERSEN, RICHARD LIN, MARTIN SCHOEBERL

February 2023

Enabling Coverage-Based Verification in Chisel

AMELIA DOBIS, HANS JAKOB DAMSGAARD, ENRICO TOLOTTO, KASPER HESSE, TJARK PETERSEN, MARTIN SCHOEBERL

ETS 2022

May 2022

Towards Functional Coverage-Driven Fuzzing for Chisel Designs

AMELIA DOBIS, TJARK PETERSEN, MARTIN SCHOEBERL

WOSET 2021

Novembre 2021

ChiselVerify: An Open-Source Hardware Verification Library for Chisel and Scala

AMELIA DOBIS, TJARK PETERSEN, HANS JAKOB DAMSGAARD, KASPER HESSE, ENRICO TOLOTTO, SIMON THYE ANDERSEN, RICHARD LIN, MARTIN SCHOEBERL

NorCaS 2021

October 2021

Best paper award

Teaching

Graduate Instructor (teach lectures, design & grade exams and assignments, supervise TAs):

COS-417: Operating Systems

Spring 2026 - Princeton University

COS-217: Introduction to Programming Systems

Fall 2025 - Princeton University

252-0061-00L: Systems Programming & Computer Architecture

Fall 2021 - ETH Zürich

Teaching Assistant (help students in lab sessions, grade assignments/exam):

DTU-02155: Computer Architecture and Engineering

Fall 2020 - Technical University of Denmark

CS-209: System-on-Chip Architecture

Spring 2020 - EPFL

Awards & Fellowships

Oregon Programming Languages Summer School Full Fellowship

July 2025

Princeton Engineering Graduate Fellowship

August 2024

Cornell Bowers CIS Dean's Excellence Fellowship

February 2024

Swiss Game Award 2022 for Best Serious Game

November 2022

NorCaS 2021 - Best Paper Award

October 2021

Service

Program Committee Member: LATTE'26, DSD'23, DSD'24, DSD'25, DSD'26, VALID'23

Journal Reviewer: TODAES - 2024, 2025

Conference Co-Chair: NJPLS Dec 2025

Steering Committee: NJPLS

Skills

Languages: English (Native), French (Native), Portuguese (Native), German (B2), Japanese (N3), Danish (A2)

Programming Languages: C/C++ (+ LLVM MLIR), Python, Scala, C#, Bash, Rust, OCaml, Haskell, Java, Kotlin,...